

P&S, Spring 2023, midterm exam

Version 2



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

FIRST NAME:

LAST NAME:

ID NUMBER:

SECTION:

SIGNATURE:

ROOM NUMBER:

MIDTERM EXAMINATION

COURSE NAME:

PROBABILITY AND STATISTICS

COURSE NUMBER:

MSC3010

EXAMINATION DURATION:

1 HOUR

ADDITIONAL MATERIALS
ALLOWED TO BE USED:

SIMPLE CALCULATOR

SPECIAL INSTRUCTIONS

(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS.
(2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET,
YOU MUST PUT ONLY YOUR **ID** ON EVERY ODD PAGE
(3) For each question write your solution in details

- Final answers must be written by only blue or black, non-erasable pen.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

Do NOT fill	Q1	Q2	Q3	Q4	Q5	Q6	Total
max points	5	5	5	5	5	5	30
your score							

Question 1:

Which is more likely: obtaining at least one head in two tosses of a fair coin, or exactly two heads in four tosses of a fair coin? Why?

two tosses: $\{HH, HT, TH, TT\}$

$$P(\text{at least one head}) = P(\{HH, HT, TH\}) = \frac{3}{4}$$

four tosses: $\{HHHH, HHHT, HHTH, HTHH, THHH, HHTT, HTHT, THTH, THTH, TTHT, HTTT, TTTT\}$

$$P(\text{exactly two heads}) = 6 \cdot \frac{1}{16} = \frac{3}{8}$$

OR another way to solve 2nd part:

$$p = \frac{C_2^4}{2^4} = \frac{4!}{2!2!} \cdot \frac{1}{2^4} = \frac{6}{16} = \frac{3}{8}$$

OR using binomial distribution:

X is number of heads, $p = \frac{1}{2}$, $n = 4$

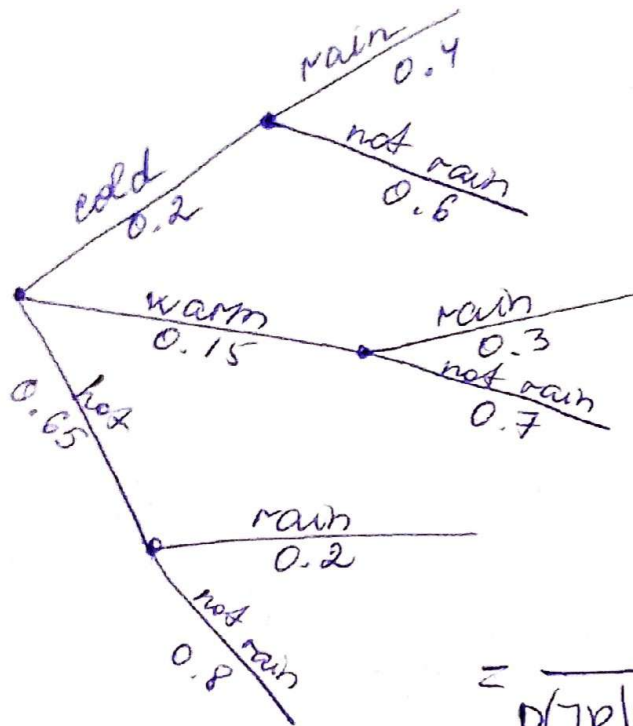
$$\Rightarrow p = P(X=2) = C_2^4 \cdot \left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^2 = \frac{3}{8}$$

Anyway $\frac{3}{8} < \frac{3}{4}$ so

it's more likely to obtain at least one head in two tosses than exactly two heads in four tosses

Question 2:

The weather on a particular day is classified as either cold, warm, or hot. There is a probability of 0.20 that it is cold and a probability of 0.15 that it is warm. In addition, on each day it may either rain or not rain. On cold days there is a probability of 0.40 that it will rain, on warm days there is a probability of 0.30 that it will rain, and on hot days there is a probability of 0.20 that it will rain. If it is not raining on a particular day, what is the probability that it is warm?



$$p(W | \text{not rain}) = ?$$

$$= \frac{p(W \cap \neg R)}{p(\neg R)} = \frac{p(\neg R | W) p(W)}{p(\neg R)}$$

$$= \frac{p(\neg R | W) p(W)}{p(\neg R | W) p(W) + p(\neg R | C) p(C) + p(\neg R | H) p(H)}$$

$$= \frac{0.7 \cdot 0.15}{0.7 \cdot 0.15 + 0.6 \cdot 0.2 + 0.8 \cdot 0.65} = \frac{0.105}{0.105 + 0.12 + 0.52} = \frac{0.105}{0.745} = \frac{105}{745} \approx 0.1409$$

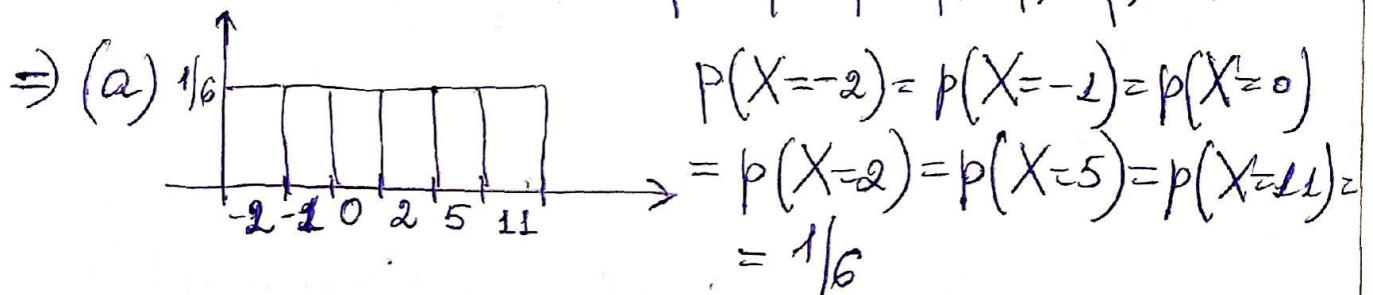
Question 3:

In a game a player rolls a fair six-sided die. If the score is even, the player receives an amount of dollars equal to the score. If the score is odd, the player receives an amount of dollars equal to the score multiplied by three. It costs the player \$4 to play the game.

- (a) What is the probability mass function of the player's net winnings (that is, winnings minus the \$4 payment)?
 (b) Find expected value of the net winning.
 (c) Find variance of the net winning.

even - score Let X be net winning
 odd - $3 * \text{score}$
 cost - $4\$$

score	1	2	3	4	5	6
X	-1	-2	5	0	11	2
p	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$



$$(b) E(X) = \frac{1}{6}(-2 - 1 + 0 + 2 + 5 + 11) = \frac{15}{6} = \frac{5}{2}$$

$$(c) \text{Var}(X) = E(X^2) - E(X)^2 = \frac{1}{6}(1 + 4 + 0 + 4 + 25 + 121) - \frac{25}{4} = \frac{155}{6} - \frac{25}{4} = \frac{310 - 75}{12} = \frac{235}{12}$$

Question 4:

Suppose that the random variables X and Y have a joint probability density function

$$f(x, y) = x(1 - y)$$

for $0 \leq x \leq 1$ and $-1 \leq y \leq 0$.

(a) What is the marginal probability density function of X ?

(b) Are the random variables X and Y independent? Why?

(c) What is $\text{Cov}(X, Y)$?

$$(a) f_X(x) = \int_{-1}^0 x(1-y) dy = x \left[y - \frac{y^2}{2} \right]_{-1}^0 = x \left[0 - \left(-1 - \frac{1}{2} \right) \right] =$$

(b) To check independence we need to check that $f(x, y) = f_X(x) f_Y(y)$ for all x, y .
So we need

$$f_Y(y) = \int_0^1 x(1-y) dx = (1-y) \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}(1-y)$$

$$f_X(x) f_Y(y) = \frac{3}{2}x \cdot \frac{1}{2}(1-y) = \frac{3}{4}x(1-y) \neq f(x, y) \Rightarrow$$

$\Rightarrow$ not independent.

$$(c) \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$E(X) = \int_0^1 x \cdot \frac{3}{2}x dx = \frac{3}{2} \cdot \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{2}$$

$$E(Y) = \int_{-1}^0 y \cdot \frac{1}{2}(1-y) dy = \frac{1}{2} \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_{-1}^0 = \frac{1}{2} \left(0 - \left(\frac{1}{2} - \frac{1}{3} \right) \right) =$$

$$E(XY) = \int_0^1 \int_{-1}^0 xy \cdot x(1-y) dy dx =$$

$$= \int_0^1 x^2 dx \cdot \int_{-1}^0 y(1-y) dy = \left[\frac{x^3}{3} \right]_0^1 \cdot \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_{-1}^0 = \frac{1}{3} \left(0 - \left(\frac{1}{2} - \frac{1}{3} \right) \right) = -\frac{5}{18}$$

$$\text{Cov}(X, Y) = -\frac{5}{18} + \frac{5}{24} = \frac{-20+15}{72} = -\frac{5}{72}$$

Question 5:

In a typical sports playoff series, two teams play a sequence of games until one team, the eventual winner, has won four games. Suppose that in each game team A beats team B with a probability of 0.6, and that the results of different games are independent.

(a) Explain how the negative binomial distribution can be used to analyze this problem.

(b) What is the probability that team A wins the series in game six?

(a) winner = win 4 games

success = "A beats B"

X is to get 4 successes in a number of games
 X

$\Rightarrow X$ is negative binomial with $p=0.6$, $r=4$

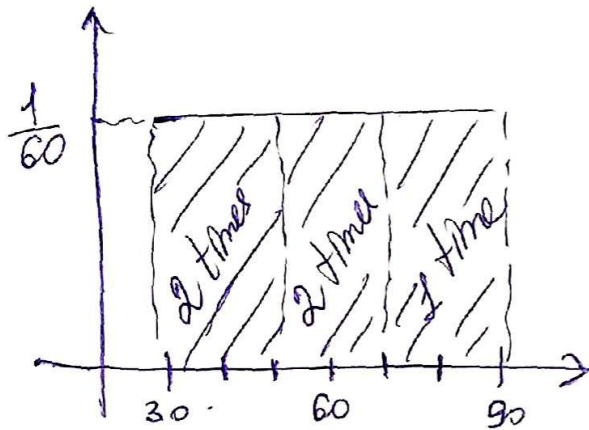
(b) $P(\text{A wins in game six}) = P(X=6) =$

$$= \binom{5}{3} \cdot 0.4^2 \cdot 0.6^4 \approx 0.2074$$

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Question 6:

Suppose that after operation, the electrical charge remaining on a circuit component has a uniform distribution between 30 and 90, and that these charges are independent of each other for different operations. If the machine is operated five times, what is the probability that the residual charge is between 30 and 50 exactly two times, between 50 and 70 exactly two times, and between 70 and 90 exactly one time?



Can be solved
using multinomial
distribution with

$$X_1=2, X_2=2, X_3=1$$

$$n=5$$

$$p_1 = 20 \cdot \frac{1}{60} = \frac{1}{3} = p_2 = p_3$$

$$\begin{aligned} \Rightarrow P(X_1=2, X_2=2, X_3=1) &= \frac{5!}{2!2!1!} \cdot \left(\frac{1}{3}\right)^2 \cdot \left(\frac{1}{3}\right)^2 \cdot \frac{1}{3} = \\ &= 30 \cdot \frac{1}{3^5} = \frac{10}{3^4} = \frac{10}{81} \approx 0.1235 \end{aligned}$$